

ISOLATED FOUR–PORT USB 2.0 HUB

ISOUSB211DPR | USB2514B | 2 x USB–A + 2 x USB–C | 480 Mbps

Schematic and placed PCB. UNROUTED: Alex routes all nets. Hardware validation pending.

HOST

Host USB–C input and current qualification

File: hub–host.kicad_sch

ISOLATION

ISOUSB211 isolated 480 Mbps USB link

File: hub–isolation.kicad_sch

CONVERTER

Isolated host power and separate shield frames

File: hub–converter.kicad_sch

EXTERNAL

Qualified external 5 V input and voltage regulation

File: hub–external.kicad_sch

DISTRIBUTION

Supply selection and protected port distribution

File: hub–distribution.kicad_sch

BUS START

Bus startup and aggregate fault reporting

File: hub–bus_start.kicad_sch

HUB

USB2514B hub, 3.3 V supply, clock and configuration

File: hub–hub.kicad_sch

PORTS A

Downstream USB–A ports 1 and 2

File: hub–ports_a.kicad_sch

PORT C3

Downstream USB–C port 3

File: hub–port_c3.kicad_sch

PORT C4

Downstream USB–C port 4

File: hub–port_c4.kicad_sch

DATA PATH

J1 host -> ISOUSB211 -> USB2514B -> J3/J4 USB–A + J5/J6 USB–C

POWER MODES

Host must advertise at least 1.5 A: 300 mA shared downstream design budget.
External J2 must advertise 3 A: 2 A total / 500 mA per port, subject to the power–report conditions.
A default–current host requires qualified external power. Source changes cause re–enumeration.

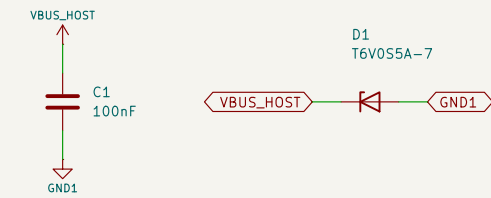
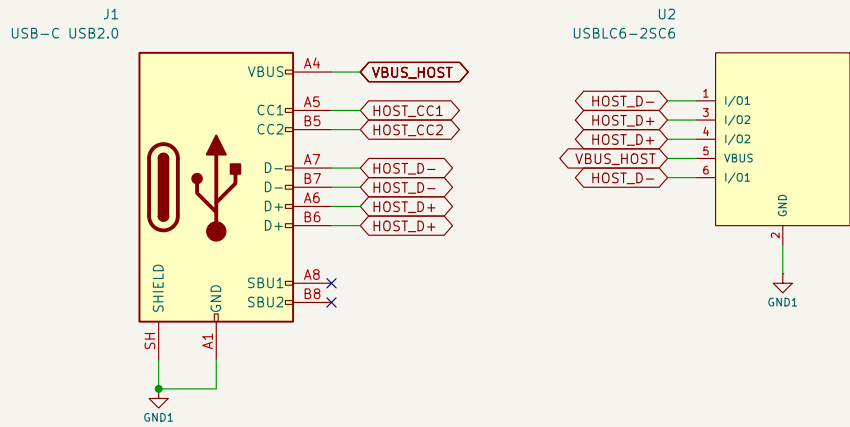
ROUTING HANDOFF

291 physical components placed; 4–layer bench board. No tracks or vias.
Separate GND1 / GND2 shield frames. TI HV (and gap: 8.20 mm nominal; other barrier rules: 8.30 mm).
Start with docs/revc/README.md for power limits, routing priorities, EEPROM and pending hardware tests.

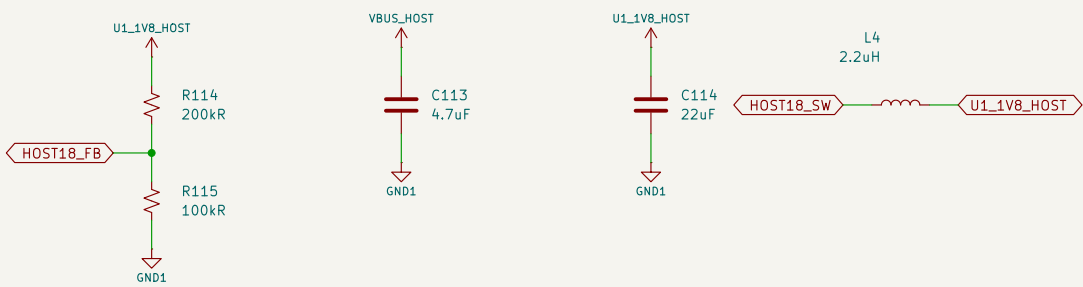
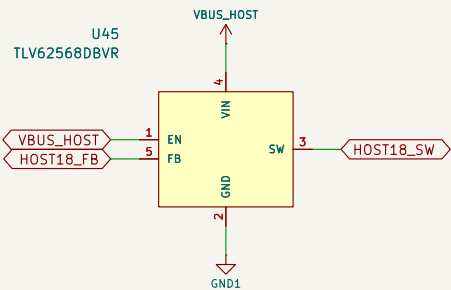
Host USB-C input and current qualification

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

Host connector and ESD



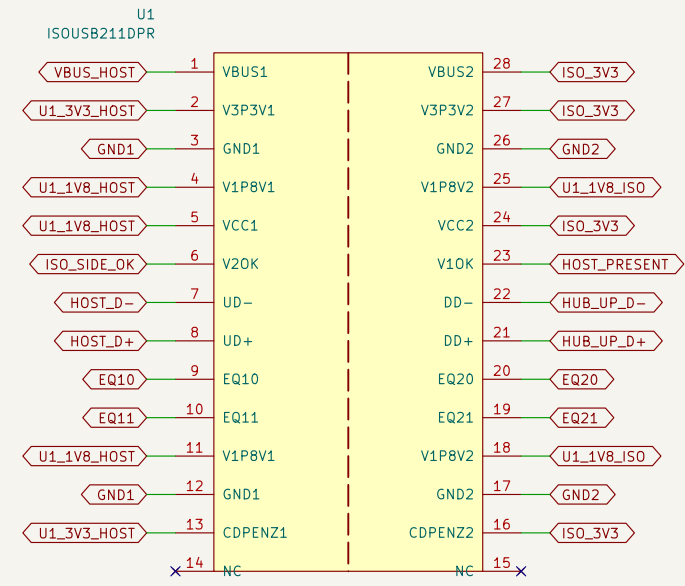
Efficient host-core supply: 1.8 V



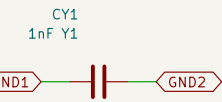
ISOUSB211 isolated 480 Mbps USB link

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

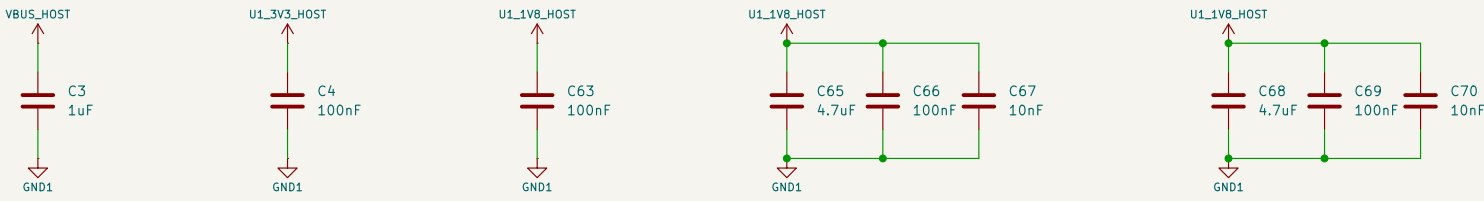
Isolated USB link: no external clock



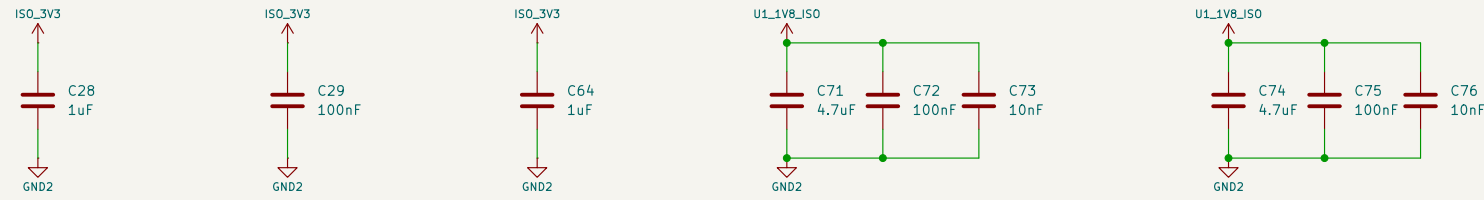
Only intentional inter-ground capacitance



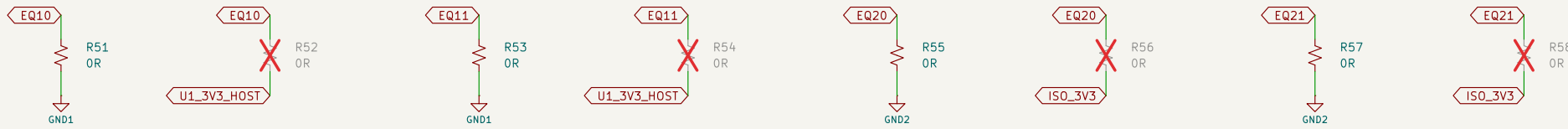
Host-side supply bypass: GND1 only



Isolated-side supply bypass: GND2 only



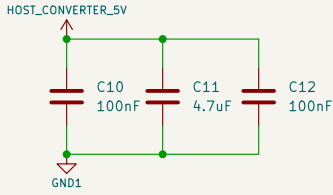
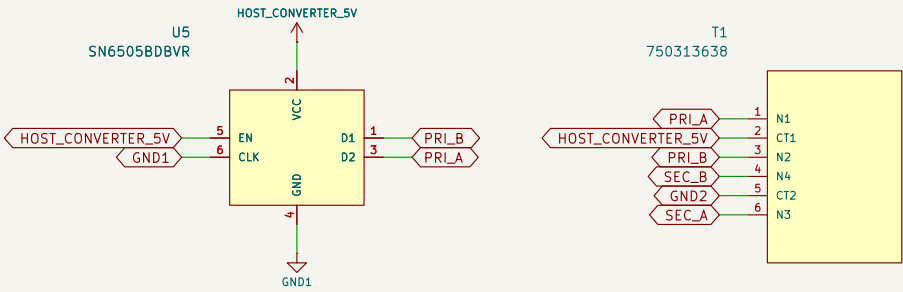
Equalization rework: fit only one strap per input



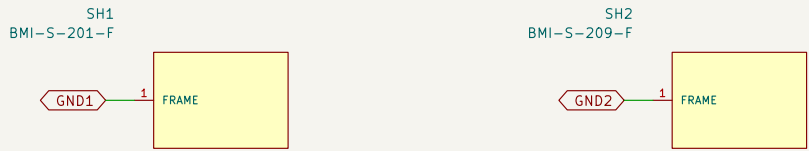
Isolated host power and separate shield frames

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

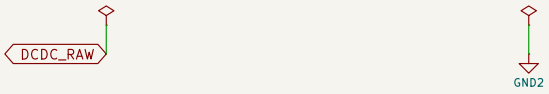
Primary push-pull driver and transformer



Two separate shield frames



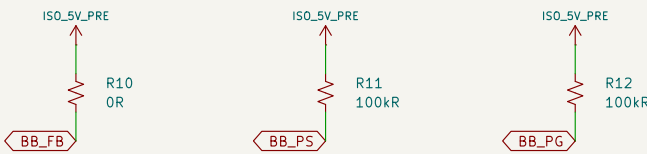
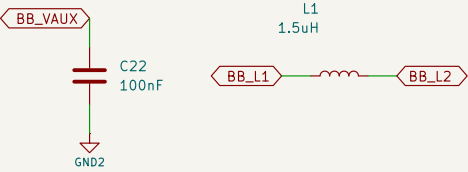
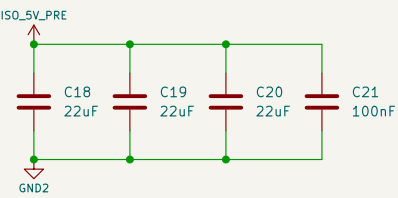
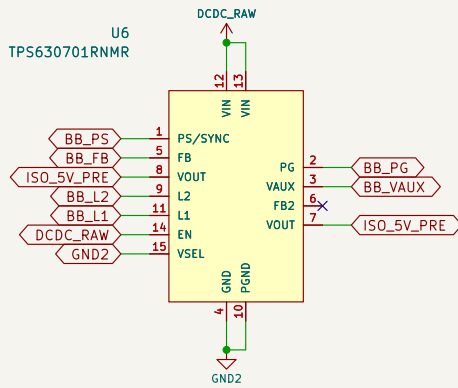
Declared power origins



Secondary rectifier and reservoir



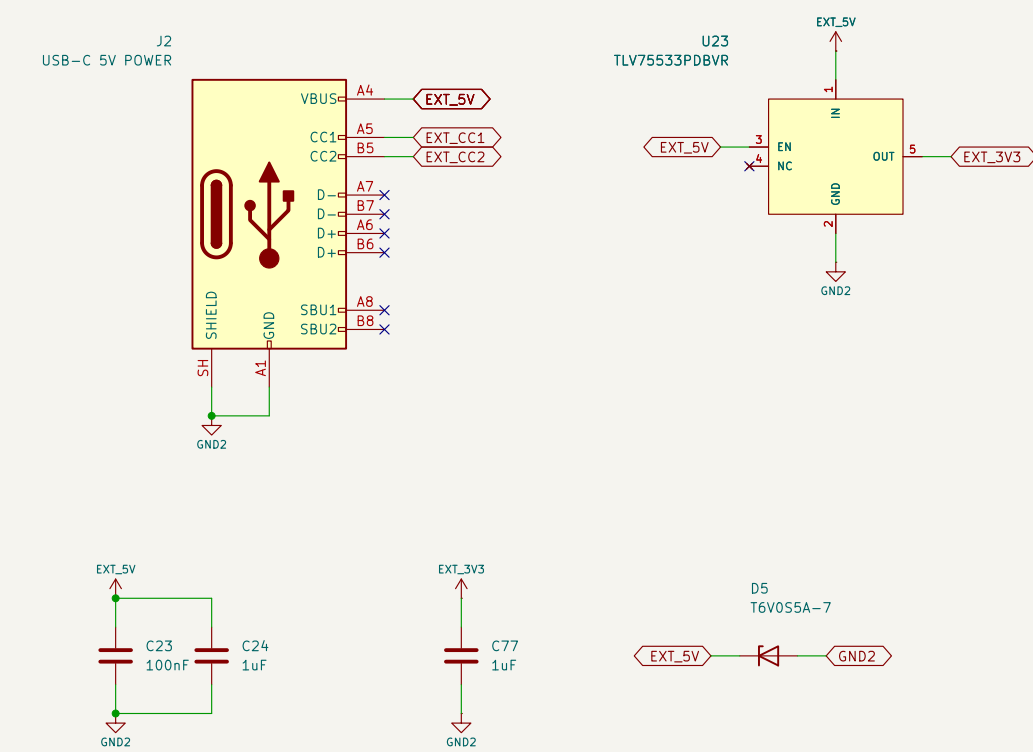
Isolated 5 V regulation



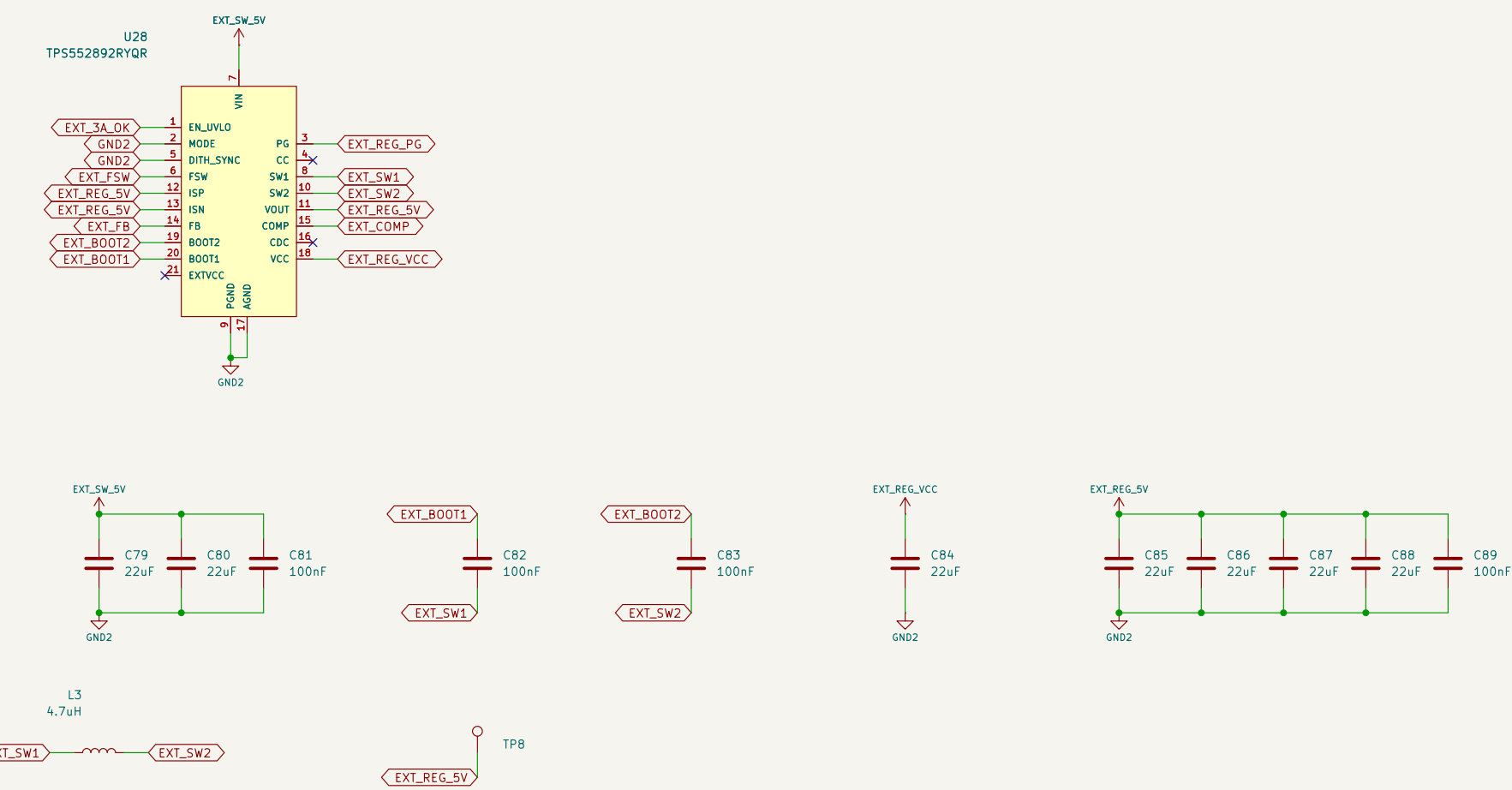
Qualified external 5 V input and voltage regulation

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

Power-only USB-C connector and 3.3 V bias



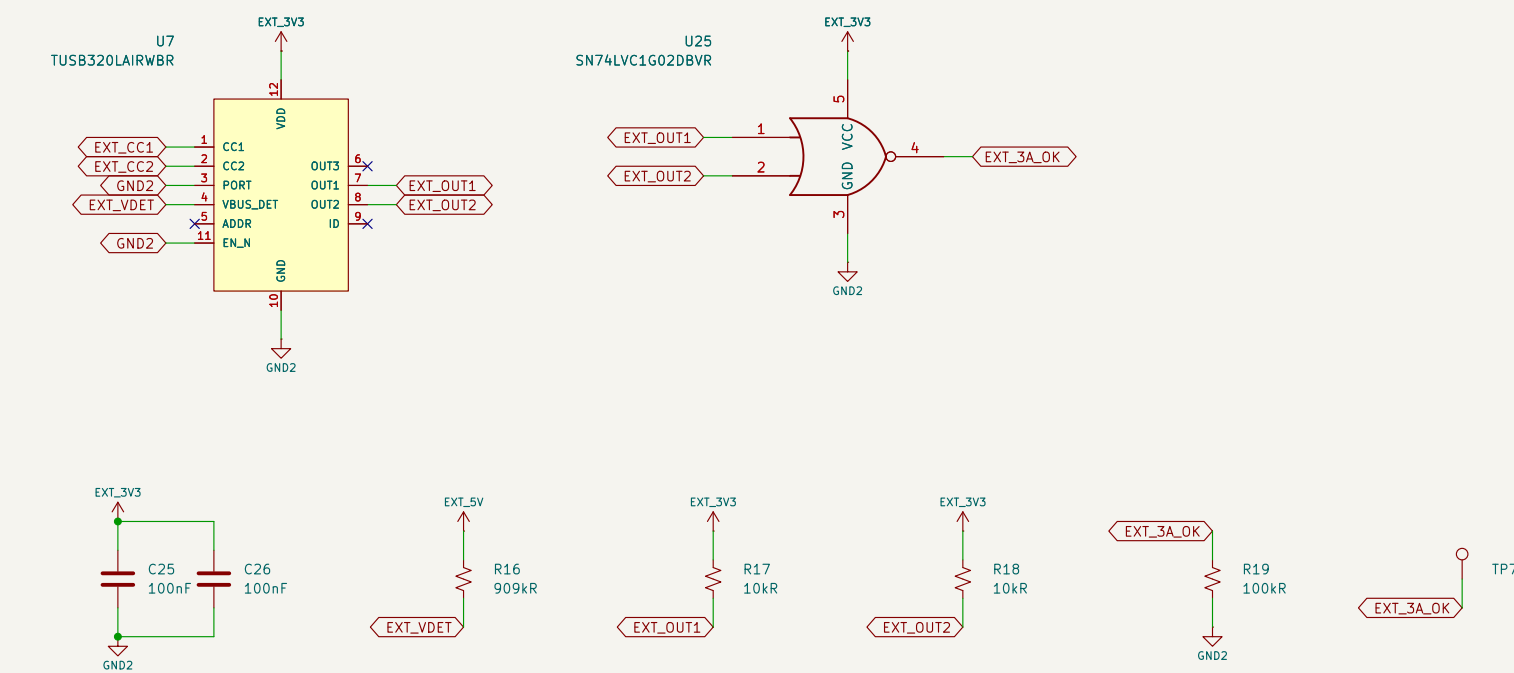
Buck–boost power stage



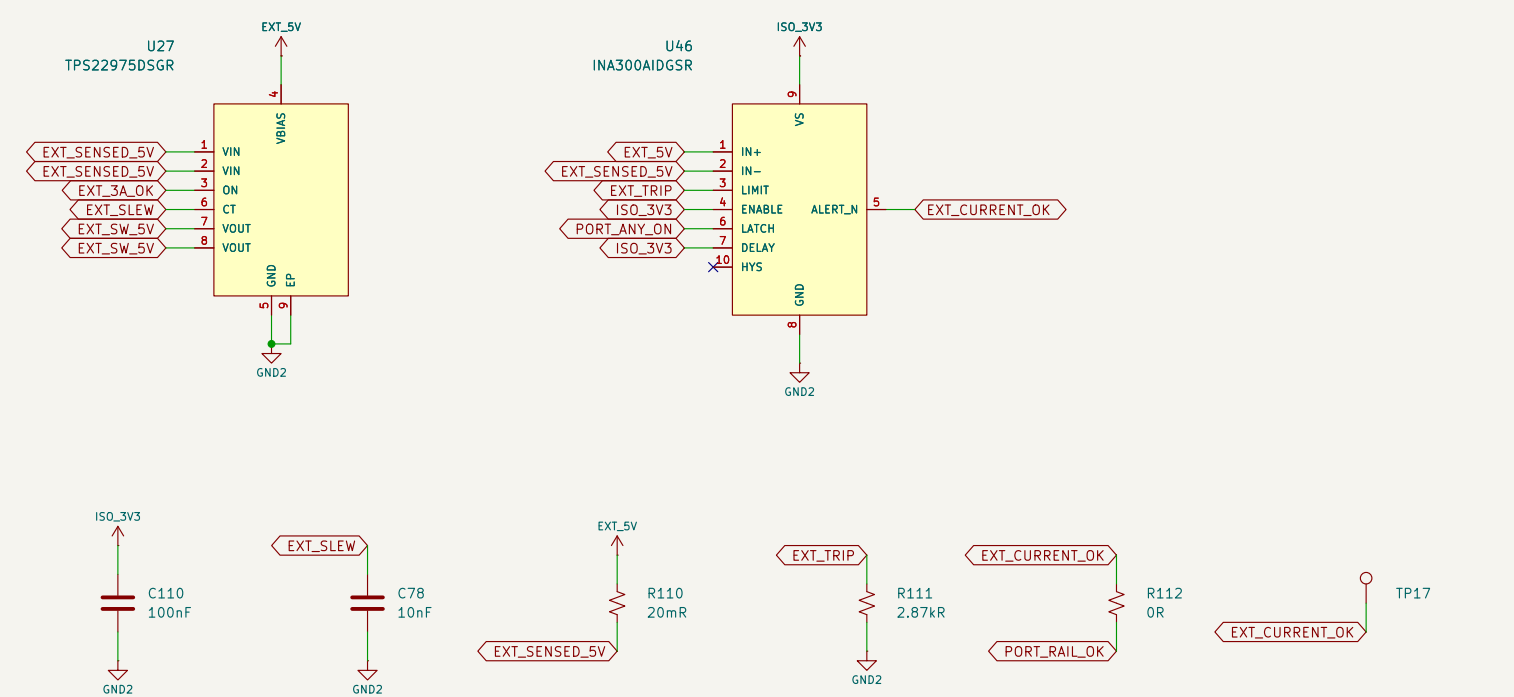
Declared power origins



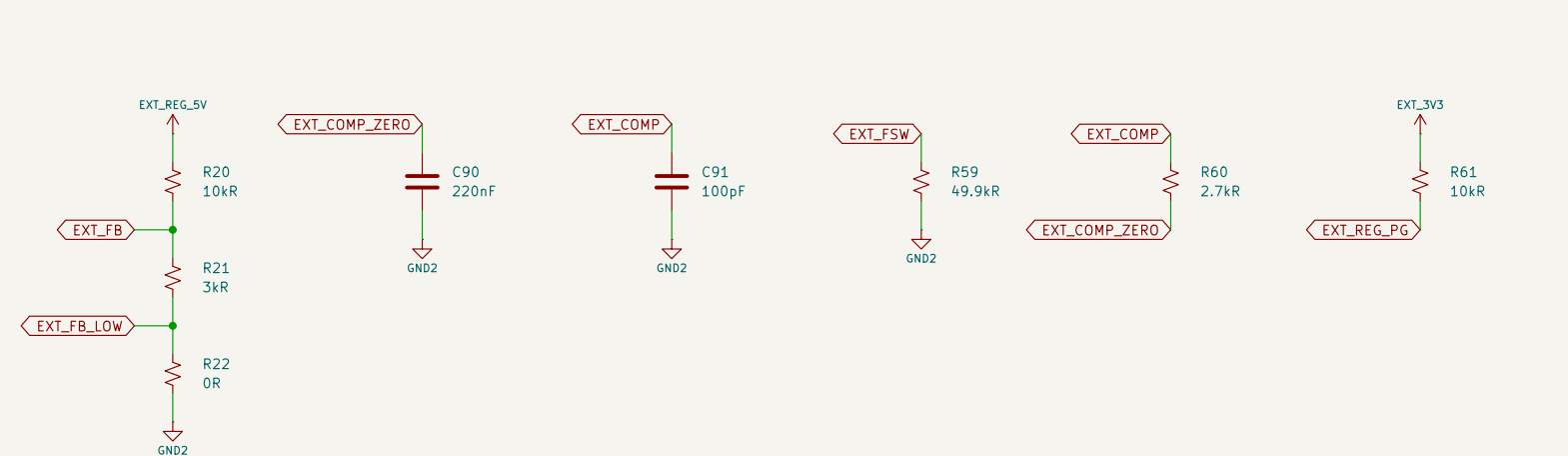
External source must advertise 3 A



Input inrush and port-shedding current monitor



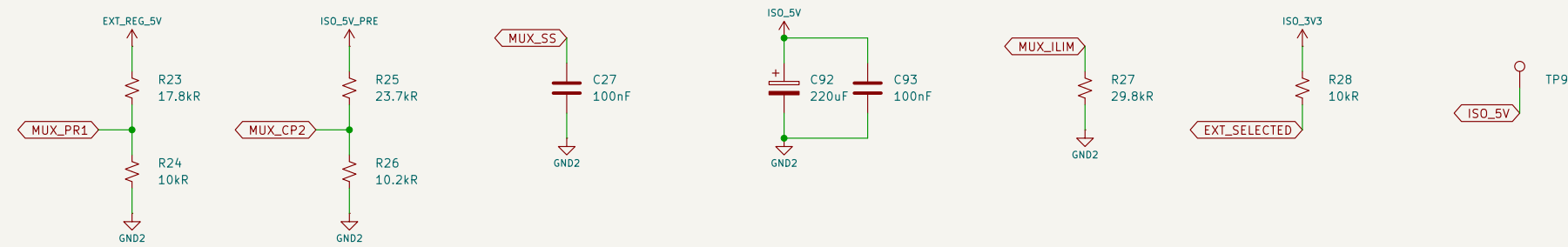
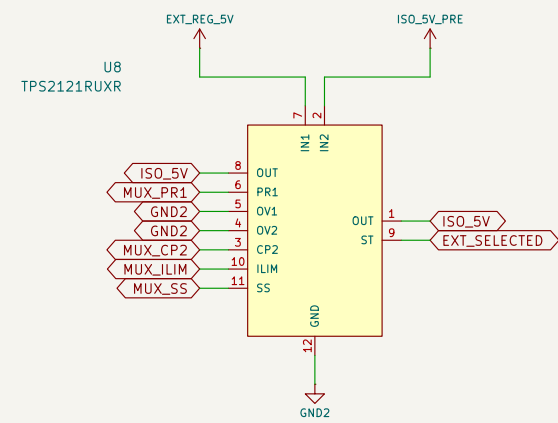
Output setpoint and voltage–loop compensation



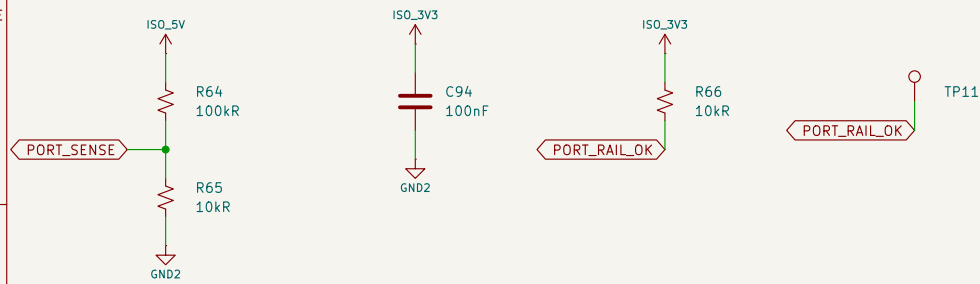
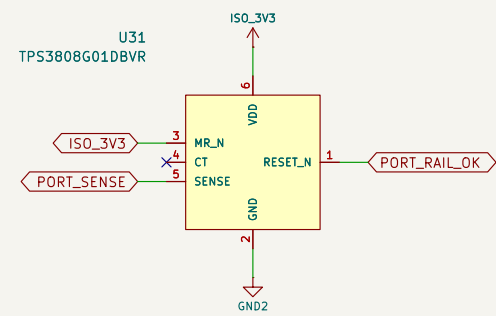
Supply selection and protected port distribution

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

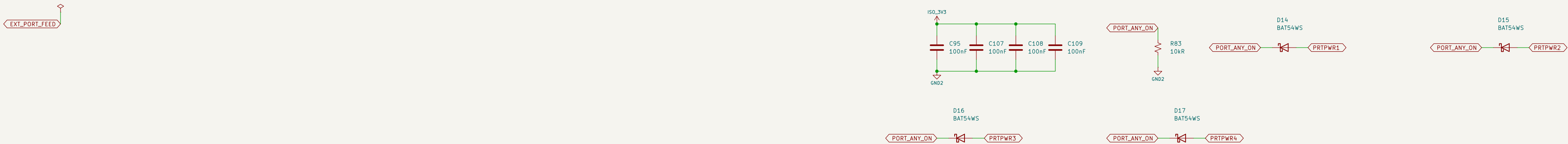
External-priority power mux and core hold-up



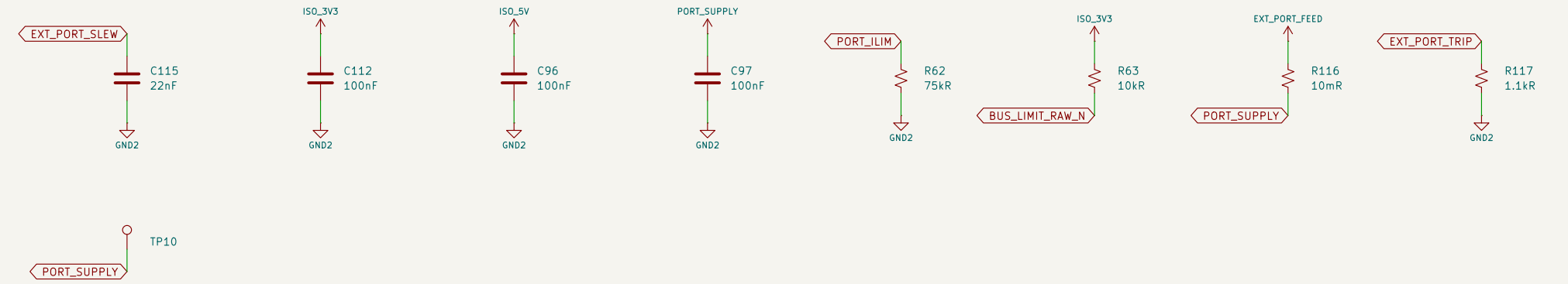
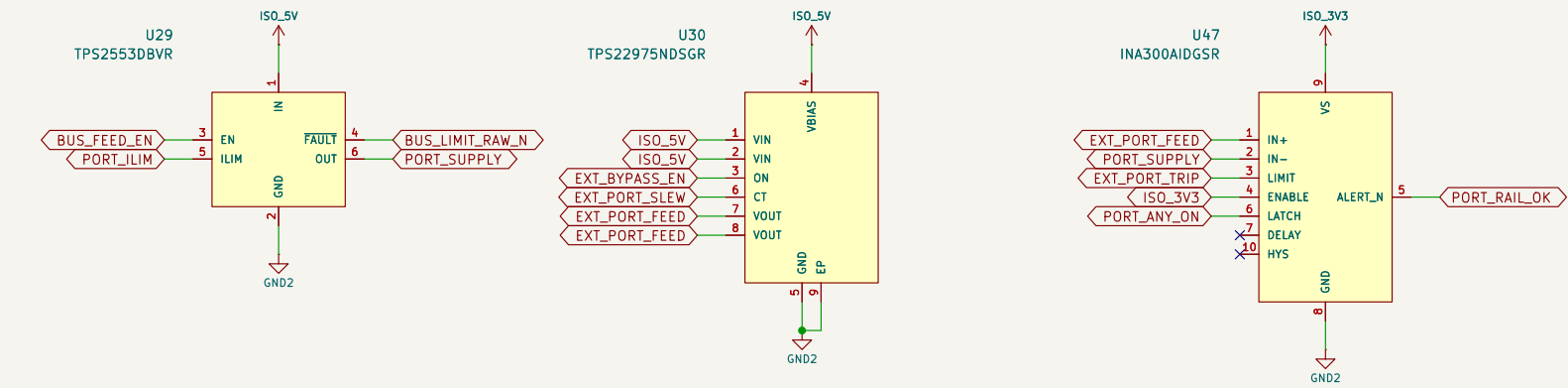
Core undervoltage sheds downstream loads



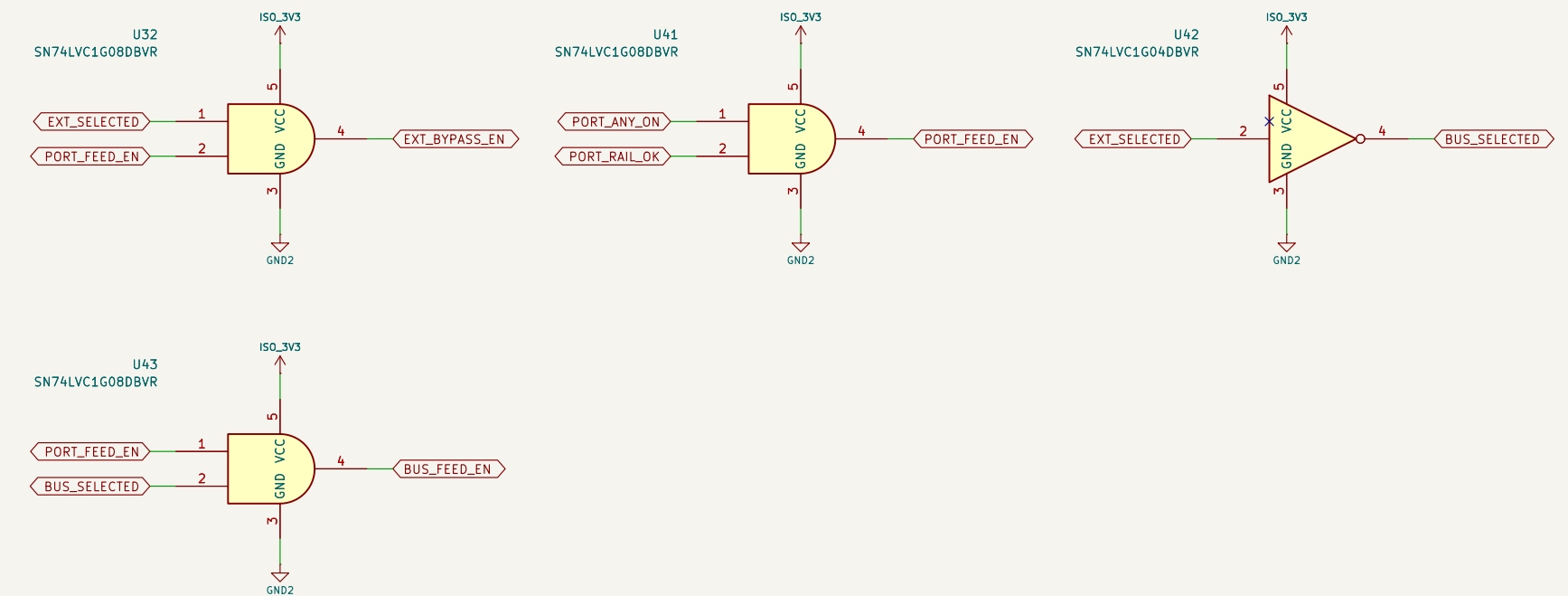
Declared power origins



Bus / external downstream supply paths



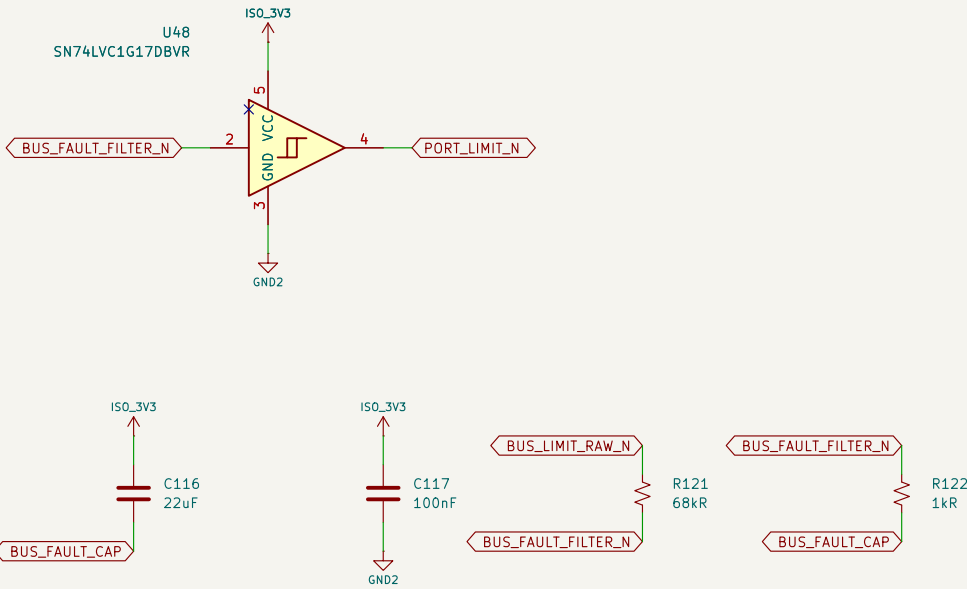
Mutually exclusive supply enables



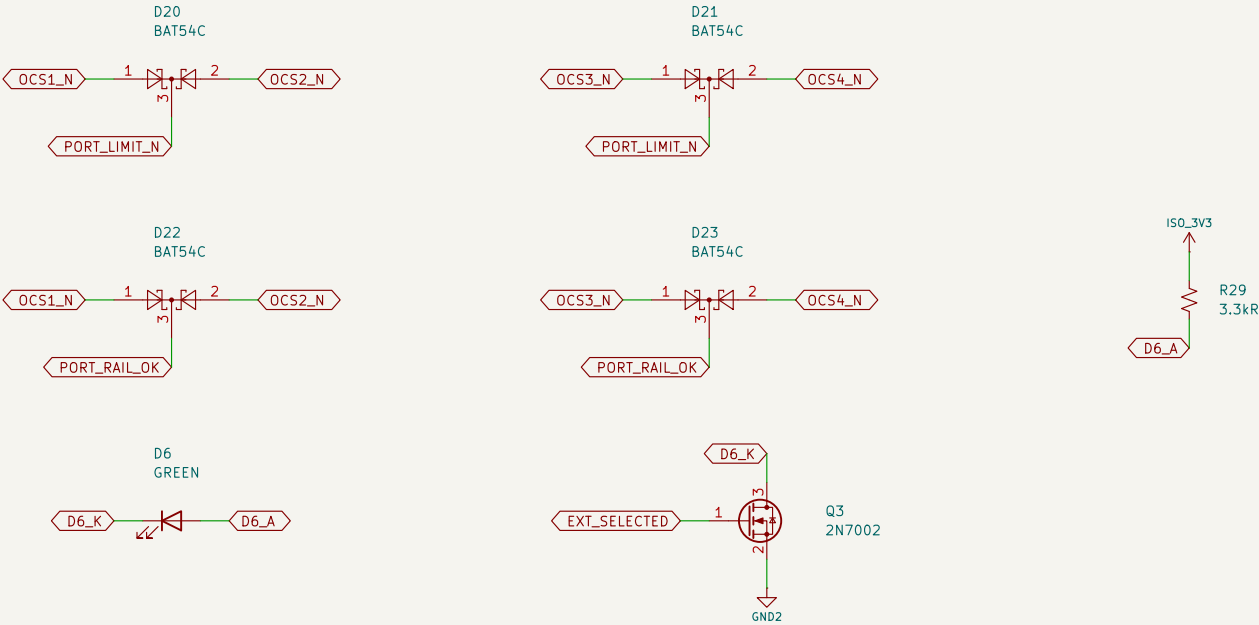
Bus startup and aggregate fault reporting

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

Delayed bus fault report; immediate current limit



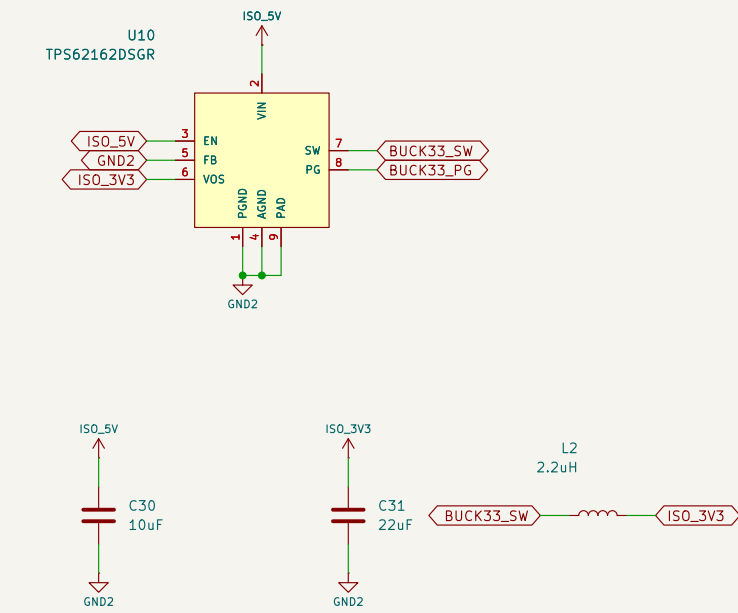
Aggregate fault fanout and external-source indicator



USB2514B hub, 3.3 V supply, clock and configuration

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

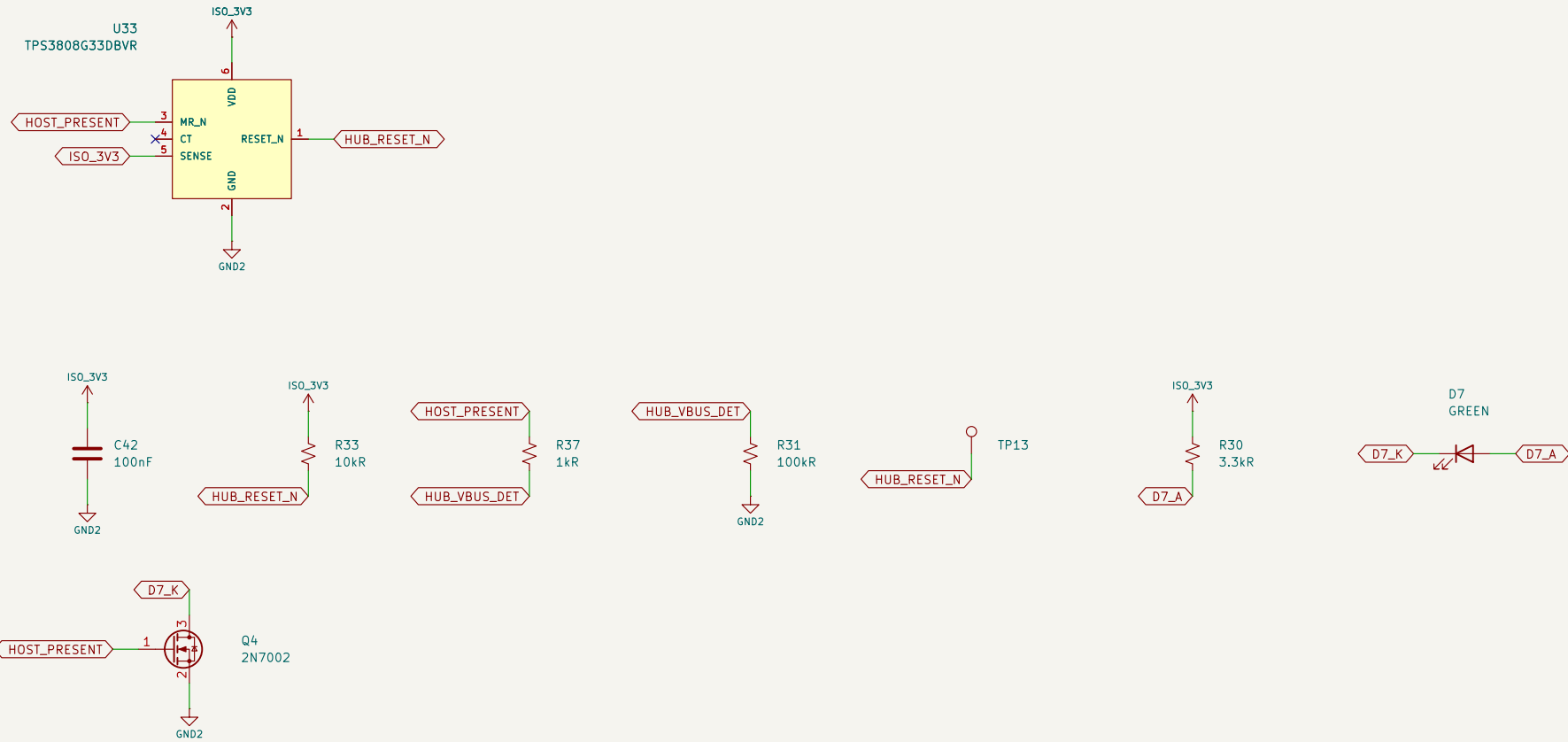
Shared 1 A rated 3.3 V buck



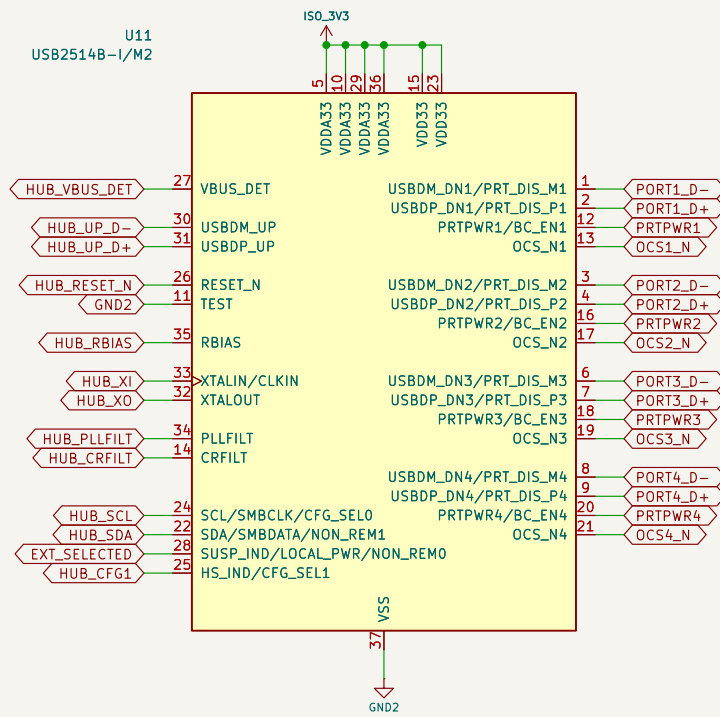
Hub clock, bias and internal-regulator filters



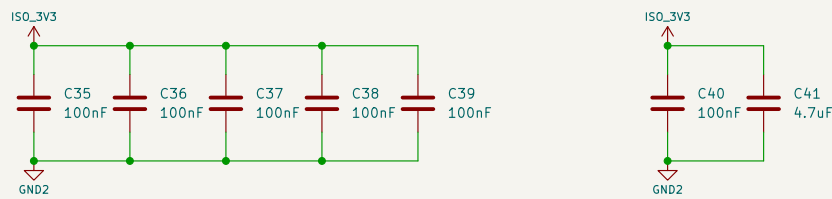
Host-present reset and VBUS sensing



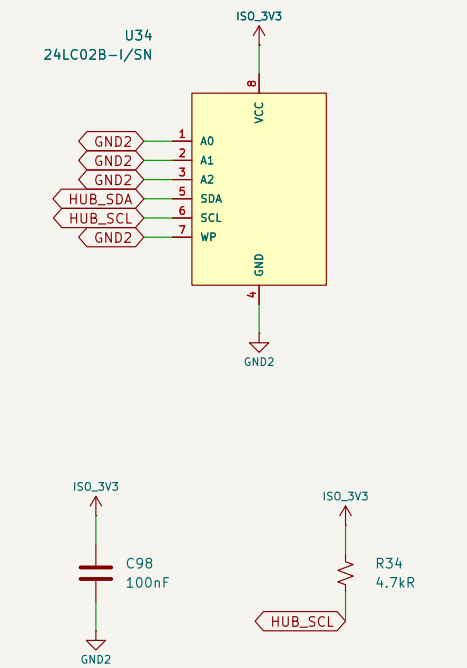
USB2514B core and per-port control



Hub local bypass: each 100 nF owns one supply pin



EEPROM configuration and programming pads



Declared power origins



Downstream USB-A ports 1 and 2

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

Port 1: USB-A, 500 mA with qualified external power

Port 2: USB-A, 500 mA with qualified external power

The diagram illustrates the electrical design for two USB-A ports, Port 1 and Port 2. Each port is designed for 500 mA with qualified external power.

Port 1 Components:

- J3 USB-A:** Connector for Port 1. Pins 1 (VBUS), 2 (D-), 3 (D+), 4 (Shield), and 5 (GND) are shown.
- U12 TPS2553DBVR:** USB-to-UART bridge. Pins 1 (EN), 2 (ILIM), 3 (FAULT), 4 (OCS1_N), and 5 (PORT1_VBUS) are shown.
- U16 USBLC6-25C6:** USB-to-USB bridge. Pins 1 (I/O1), 2 (I/O2), 3 (I/O2), 4 (I/O2), 5 (VBUS), and 6 (I/O1) are shown.
- Passive Components:** C43 (100nF), C44 (220uF), C45 (100nF), R38 (46.4kR), R47 (100kR), R101 (100kR), D8 (T6V055A-7), and D9 (T6V055A-7).

Port 2 Components:

- J4 USB-A:** Connector for Port 2. Pins 1 (VBUS), 2 (D-), 3 (D+), 4 (Shield), and 5 (GND) are shown.
- U13 TPS2553DBVR:** USB-to-UART bridge. Pins 1 (EN), 2 (ILIM), 3 (FAULT), 4 (OCS2_N), and 5 (PORT2_VBUS) are shown.
- U17 USBLC6-25C6:** USB-to-USB bridge. Pins 1 (I/O1), 2 (I/O2), 3 (I/O2), 4 (I/O2), 5 (VBUS), and 6 (I/O1) are shown.
- Passive Components:** C46 (100nF), C47 (220uF), C48 (100nF), R39 (46.4kR), R48 (100kR), R102 (100kR), D8 (T6V055A-7), and D9 (T6V055A-7).

Legend:

- PORT1_VBUS: Port 1 VBUS
- PORT1_D+: Port 1 D+
- PORT1_D-: Port 1 D-
- PORT1_ILIM: Port 1 ILIM
- PORT1_OCS1_N: Port 1 OCS1_N
- PORT2_VBUS: Port 2 VBUS
- PORT2_D+: Port 2 D+
- PORT2_D-: Port 2 D-
- PORT2_ILIM: Port 2 ILIM
- PORT2_OCS2_N: Port 2 OCS2_N
- ISO_3V3: 3V3
- PRT_PWR1: PRT_PWR1
- PRT_PWR2: PRT_PWR2
- GND2: GND2

Sheet: /PORTS A/
File: hub-ports_a.kicad_sch
Title: Downstream USB-A ports 1 and 2
Size: A3 | Date: 2026-09-10 | Rev: C
KiCad E.D.A. 10.99.0-3770-gd7e34de179 | Id: 9/11

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Downstream USB-A ports 1 and 2

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

Port 1: USB-A, 500 mA with qualified external power

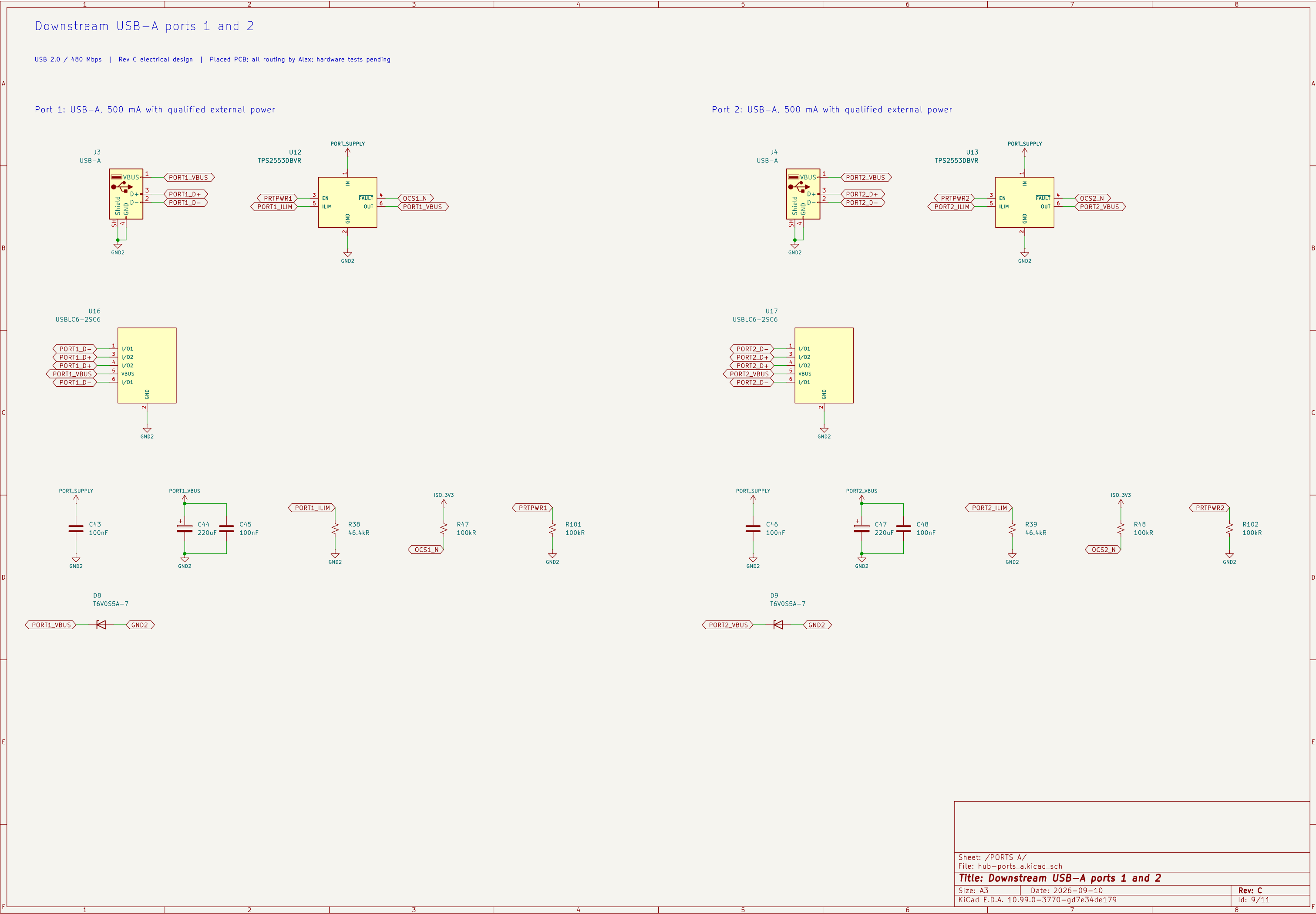
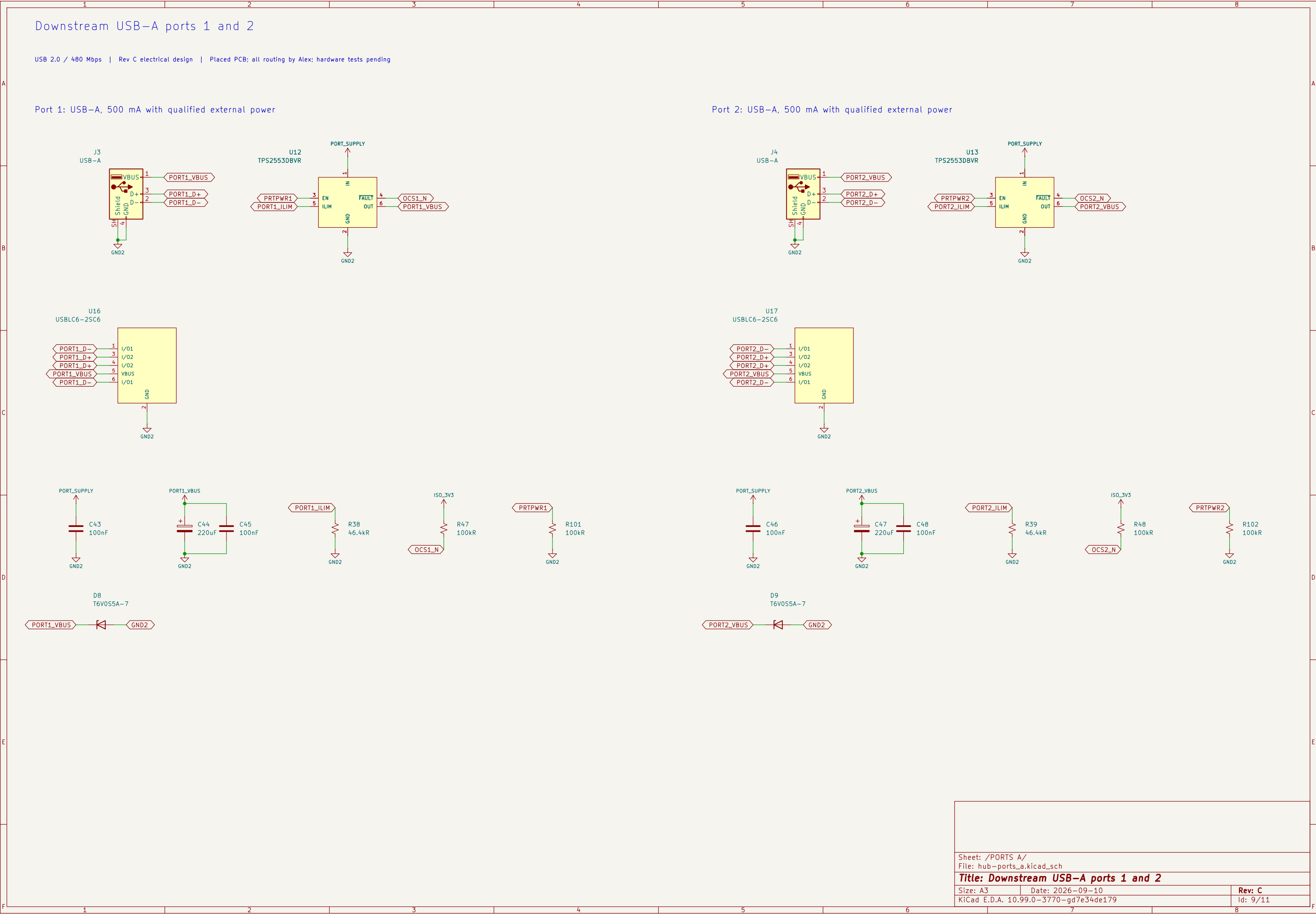
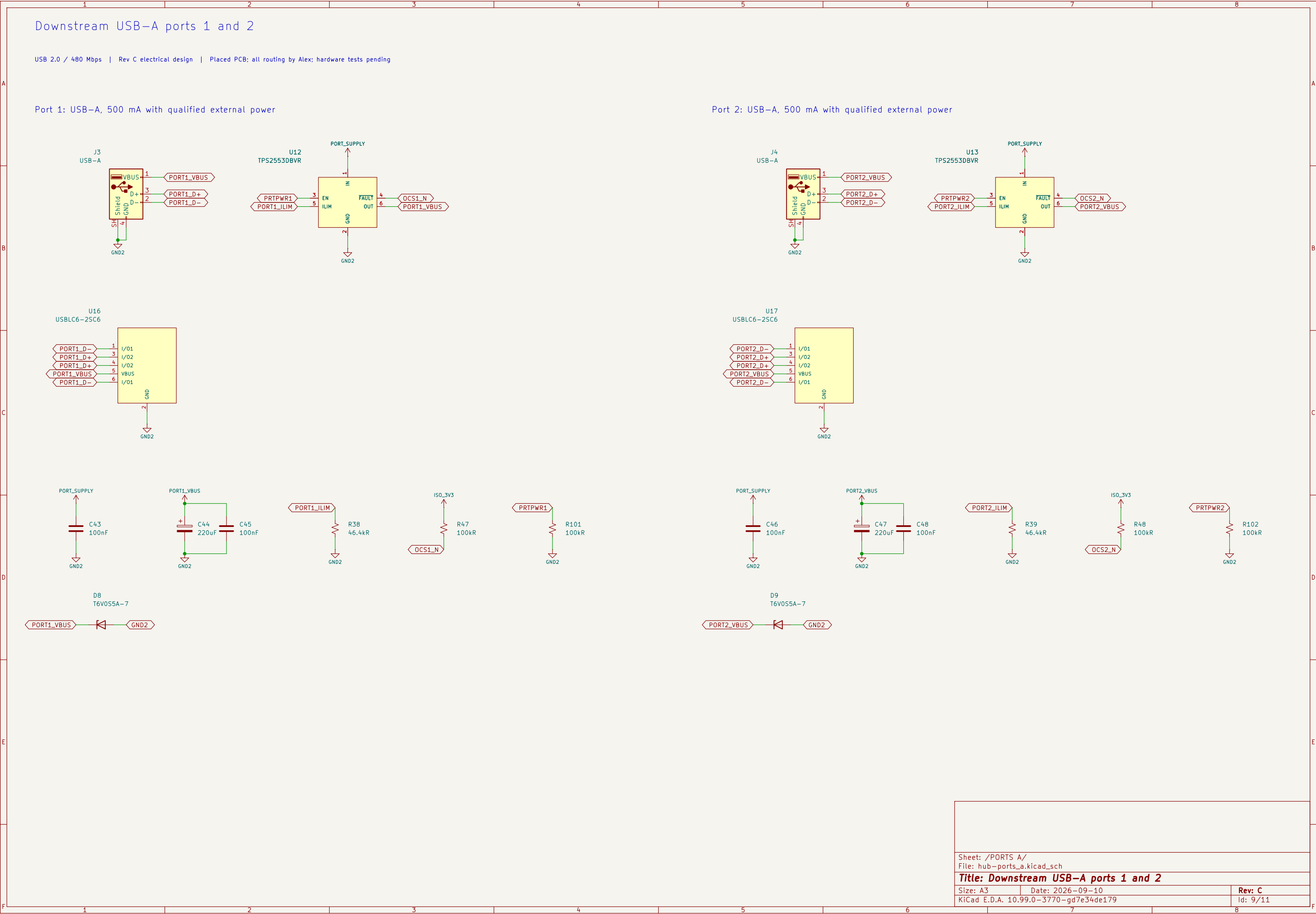
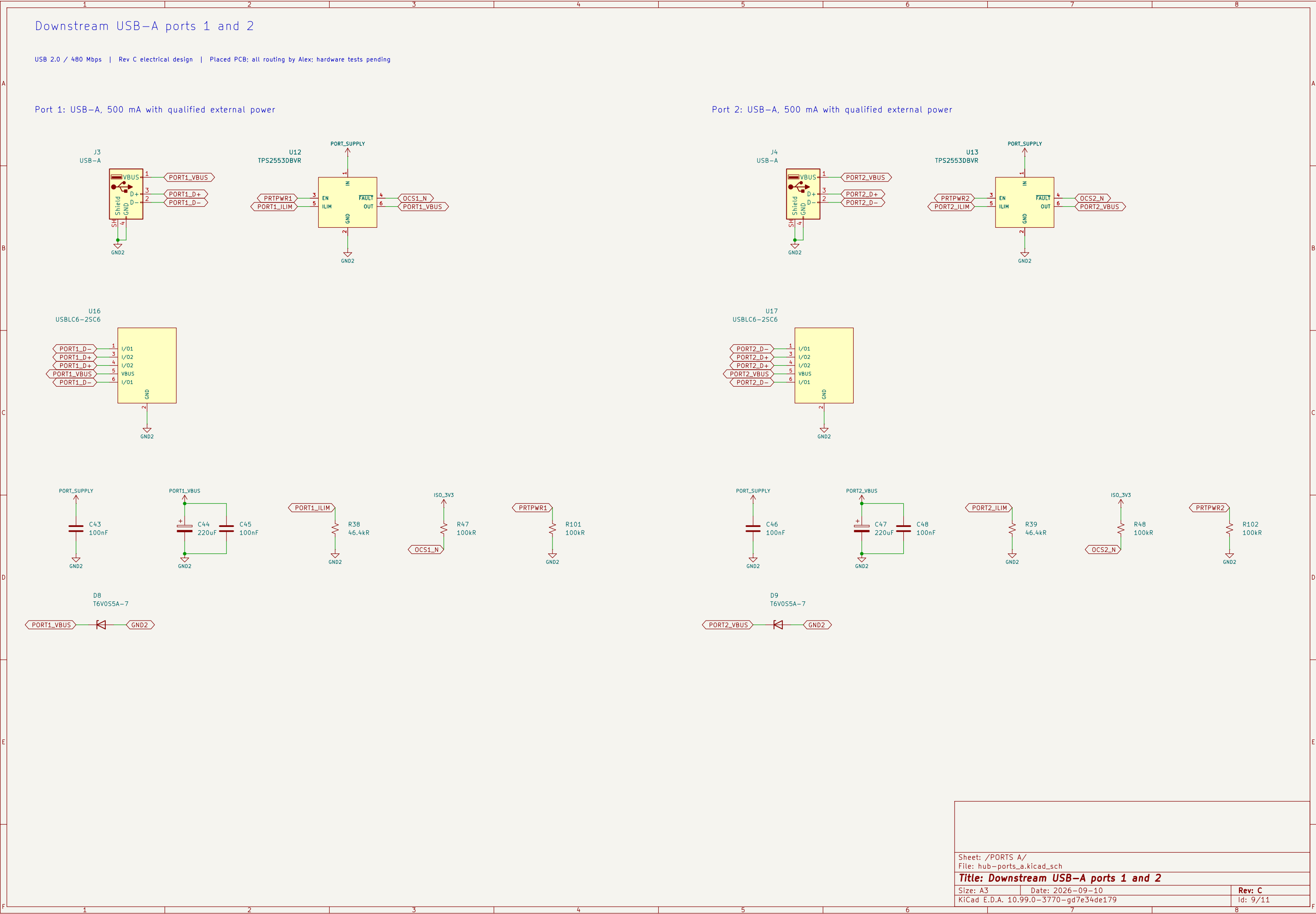
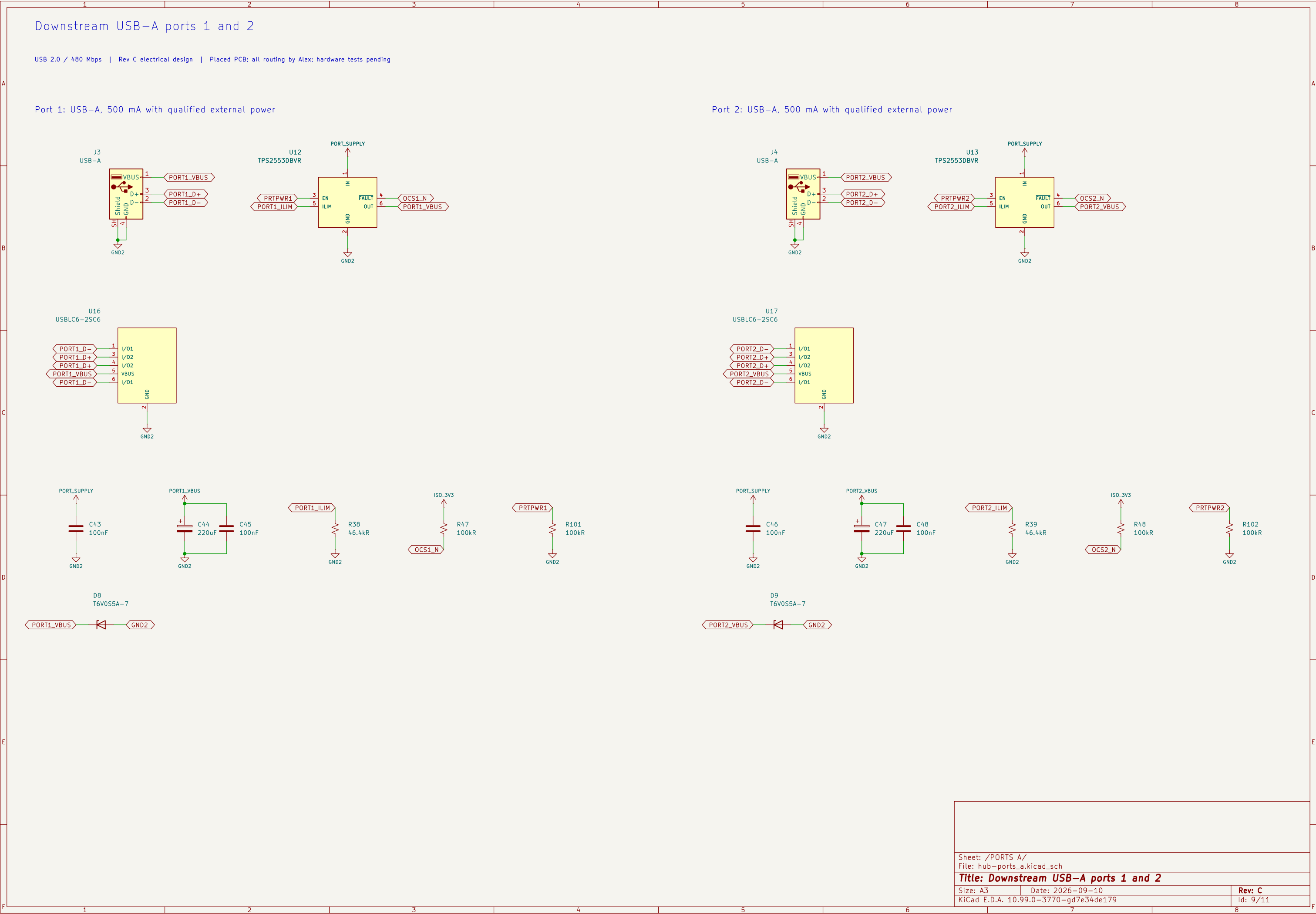
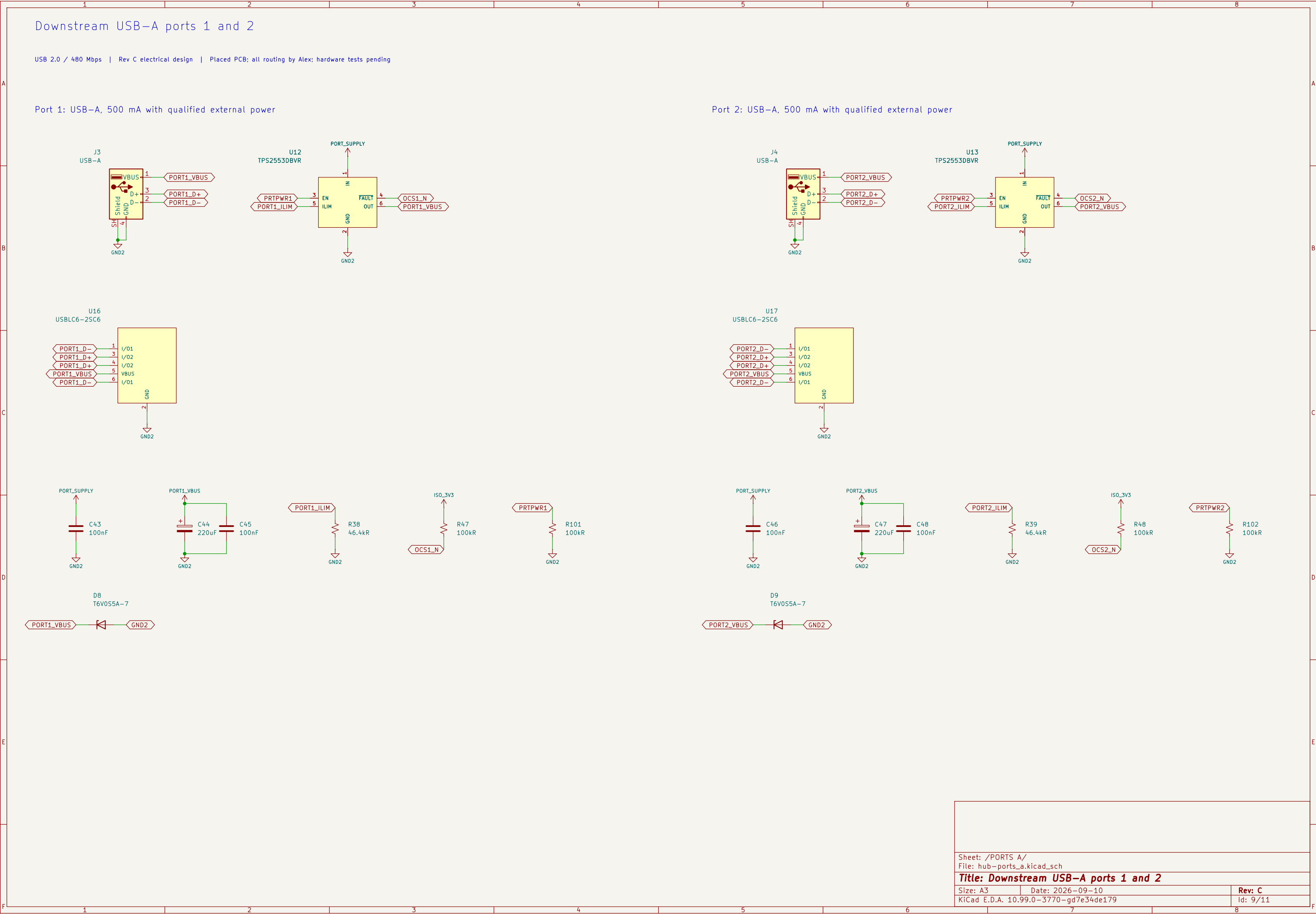
Port 2: USB-A, 500 mA with qualified external power

The diagram illustrates the electrical design for two USB-A ports, Port 1 and Port 2. Each port is powered by a 500 mA external supply (PORT_SUPPLY) and includes a USB-to-UART bridge (U16, U17) and a USB-to-USB bridge (U12, U13). The power section for each port includes a 100 nF capacitor (C43, C46), a 220 uF capacitor (C44, C47), and a 100 nF capacitor (C45, C48). The data section includes a 46.4 kΩ resistor (R38, R39) and a 100 kΩ resistor (R47, R48). The control section includes a 100 kΩ resistor (R101, R102) and a 100 kΩ resistor (R48, R102). The USB-to-UART bridge (U16, U17) is connected to the USB connector (J3, J4) and the USB-to-USB bridge (U12, U13). The USB-to-USB bridge (U12, U13) is connected to the USB connector (J3, J4) and the USB-to-UART bridge (U16, U17). The USB connector (J3, J4) is connected to the USB-to-UART bridge (U16, U17) and the USB-to-USB bridge (U12, U13). The USB-to-UART bridge (U16, U17) is connected to the USB connector (J3, J4) and the USB-to-USB bridge (U12, U13). The USB-to-USB bridge (U12, U13) is connected to the USB connector (J3, J4) and the USB-to-UART bridge (U16, U17).

Sheet: /PORTS A/
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Title: Downstream USB-A ports 1 and 2

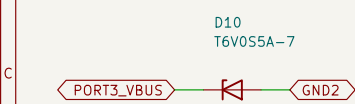
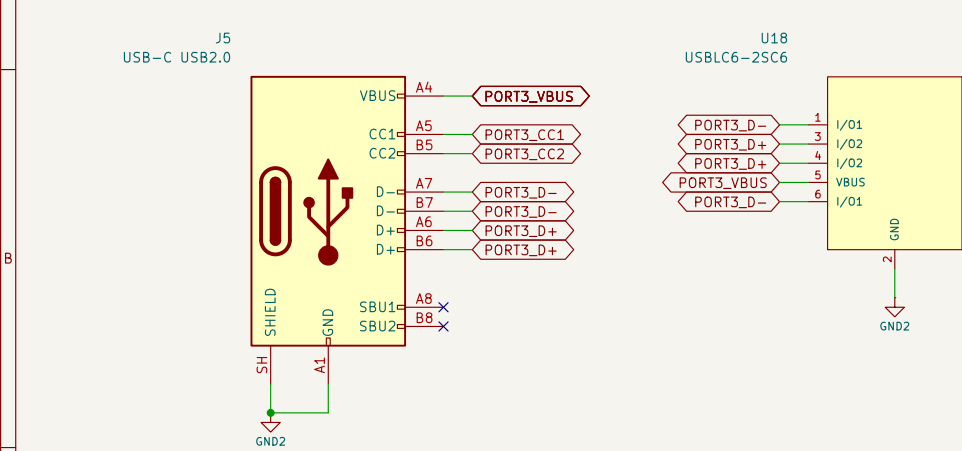
Size: A3 | Date: 2026-09-10 | Rev: C
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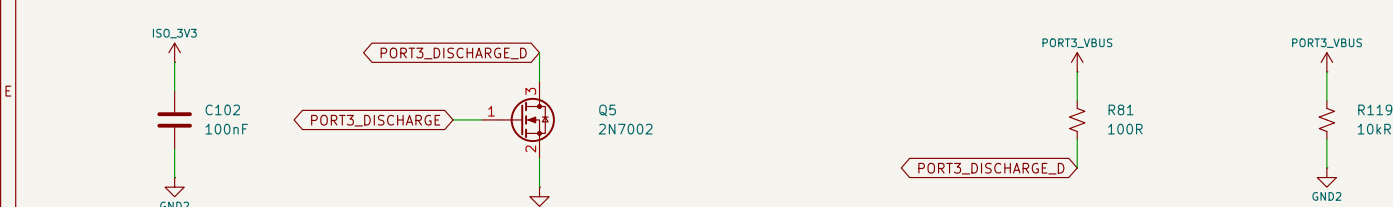
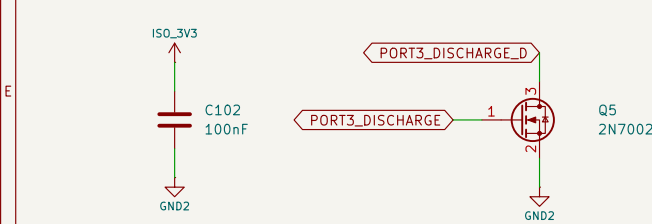
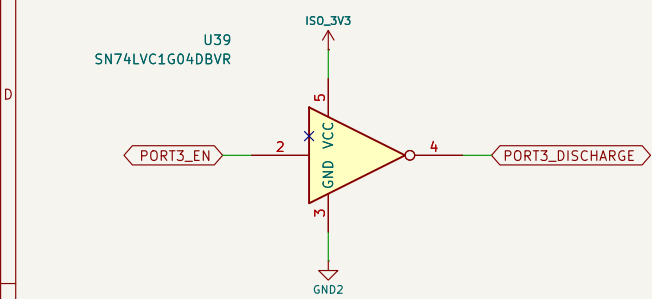
Downstream USB-C port 3

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

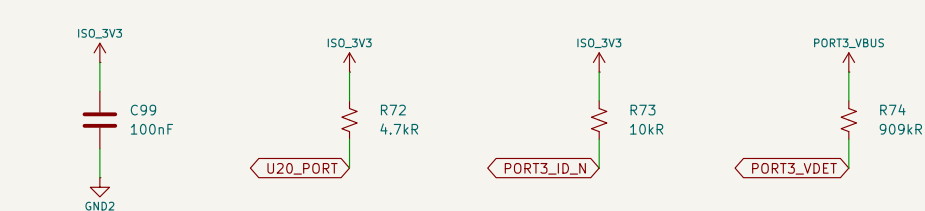
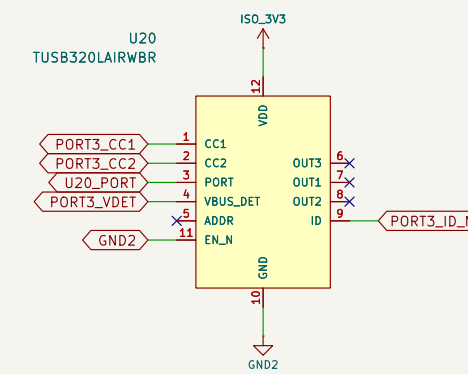
Port 3: USB-C, USB 2.0 and ESD



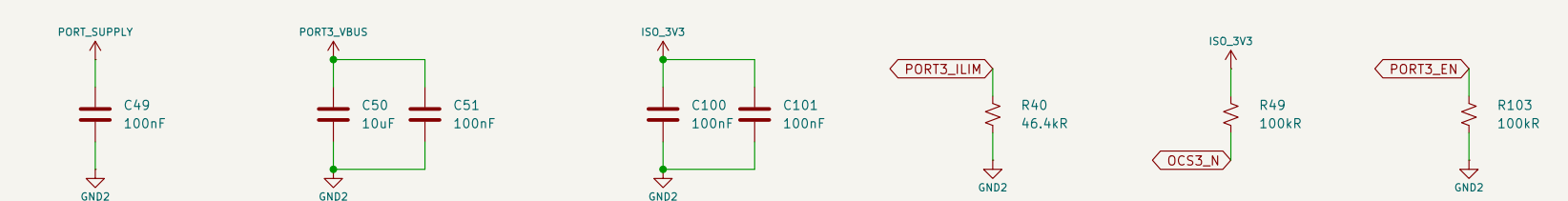
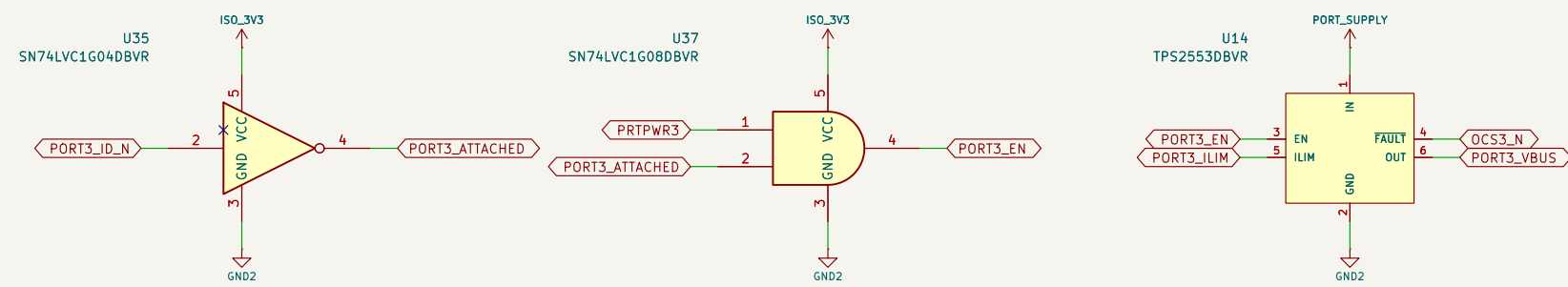
Detach discharge, including total board power loss



Attach detection: DFP, default Rp, no VCONN



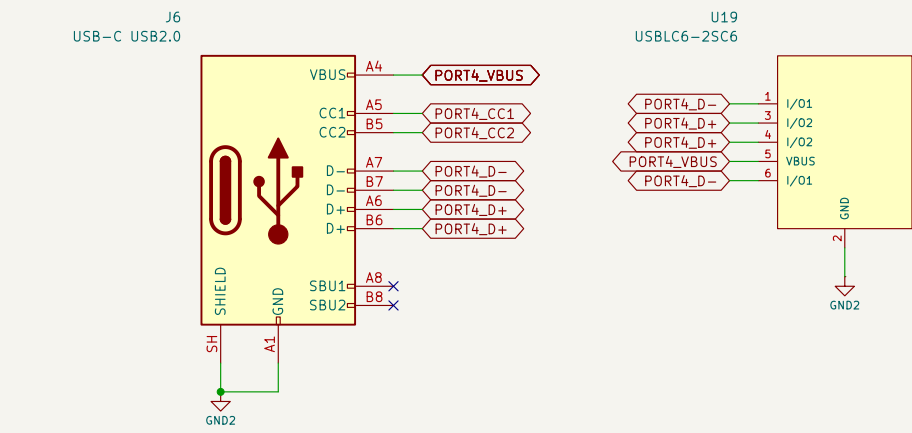
Attachment AND hub permission controls VBUS



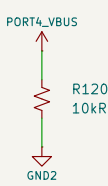
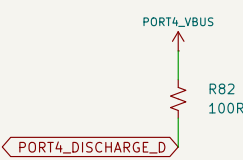
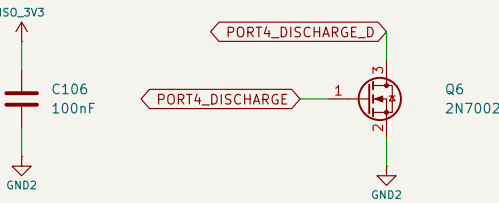
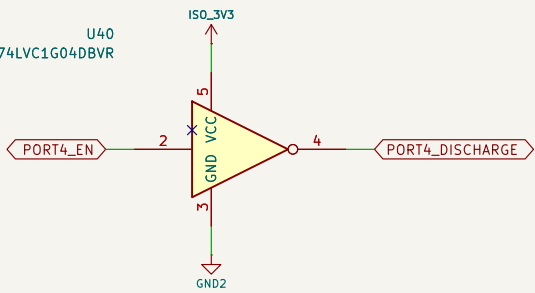
Downstream USB-C port 4

USB 2.0 / 480 Mbps | Rev C electrical design | Placed PCB; all routing by Alex; hardware tests pending

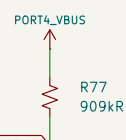
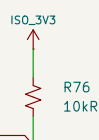
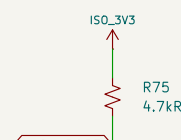
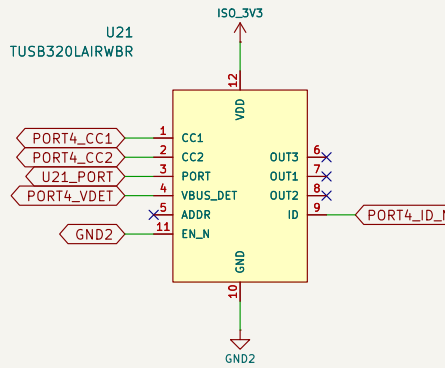
Port 4: USB-C, USB 2.0 and ESD



Detach discharge, including total board power loss



Attach detection: DFP, default Rp, no VCONN



Attachment AND hub permission controls VBUS

